Ribosome profiling with protein language models

Abstract

We measure translation with a new assay and report an AUROC of 0.87 on held-out genes.

1 Introduction

Translation is measured by sequencing ribosome-protected fragments. Earlier work used linear models only.

2 Methods

We fine-tune a protein language model on 12,000 transcripts with a contrastive loss.

3 Results

The model reaches 0.87 AUROC and beats the baseline by 9 points.

4 Discussion

Limitations remain for rare codons.

References

[1] Ingolia et al. Science 2009.